SWE5006 PRACTICE PROJECT · TEAM 8

BLOCBUY

ONE FOR ALL • ALL FOR ONE

A scalable community group-buying platform

Two prices • pay first to team up • refunded if the team doesn't fill

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Graduate Certificate in Designing Modern Software Systems · 26 October 2026

Group buying only works if enough people join in time

A market coordination failure, solved with a pay-first team purchase modelled on the early Pinduoduo playbook (proposal §4)

Today

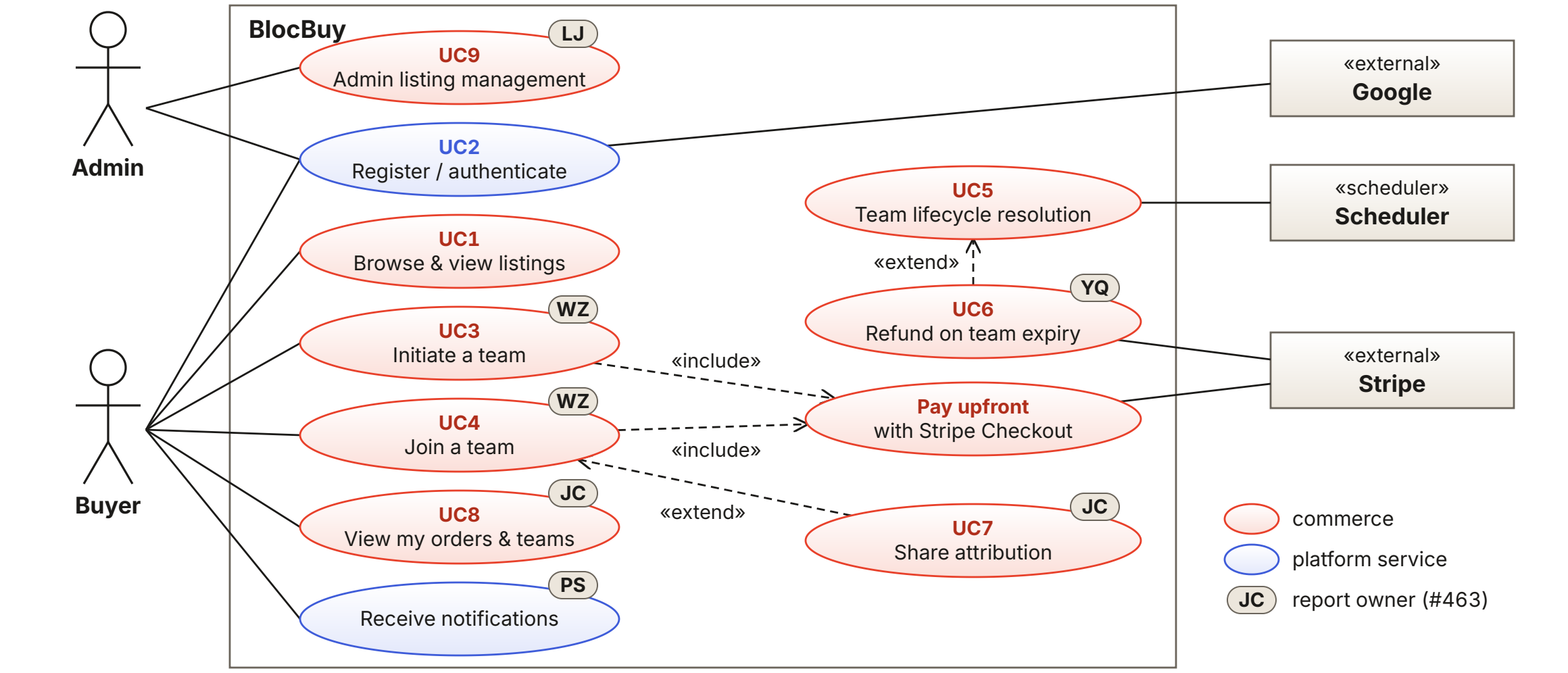
- Merchants can discount only when volume is predictable
- One buyer can never reach a bulk price
- Advertising is costly and not tied to real purchases

With BlocBuy: pay-first teams

- Two prices: individual and team
- Pay upfront to start a team, then share the invite link
- Friends join by paying upfront too
- Fills in time: team price. Expires: everyone refunded automatically

Stripe test mode only: no live money moves

Scope: nine use cases on one platform



Out of scope: merchant self-service GUI · analytics · recommendations · automatic team matching · email, SMS or push · native apps · live payments

JC Lim

Four sprints, one board, one Definition of Done

Sprint 1 — 2–12 Sep

Walking skeleton: CI/CD and dev live on 5 Sep; sign-in, browse, admin

Pipeline live; features late

35 team PRs · 3 issues closed

Sprint 2 — 14–25 Sep

One idempotent chain: payment → team → order

9 of 11 stories · 134 of 150 h

64 team PRs · 47 issues closed

Sprint 3 — 28 Sep–9 Oct

Invite → pay → refund on the deployed dev site

Not met on dev · 117 of 150 h

126 team PRs · 67 issues closed

Sprint 4 — 12–23 Oct

Complete system on dev; three services on ECS for SWE5001

`[FILL: result, 23 Oct]`

38 team PRs · 17 closed so far

Planning — plan.md, hours

Day log — 35 dated logs

Mid-sprint — check, S3–S4

Review — review, report

Retro — in the report

159 cards on the board: 132 Done, 7 in progress, 20 to do

356 PRs merged: 263 team, 70 Dependabot, 23 snapshot

Definition of Done, Sprints 3–4 (highlights)

- Acceptance criteria met; contract before controller
- Every gate green: Maven verify, ArchUnit, Gitleaks, Trivy, SonarCloud, `pnpm check`
- **"On dev" means deployed**: a local run does not count
- PR under ~1,500 lines; remaining hours on the board daily

Was it met?

- Required gates block every merge
- Sprint 3 not counted: Checkout off on dev
- Daily remaining hours not kept

Milestones and effort per member

27 Aug	5 Sep	14 Sep	22 Sep	27 Sep	8 Oct	11 Oct	15 Oct	23 Oct
Proposal	Backend live on dev	Sprint 1 tag	Web app live on dev	Sprint 2 tag	Startable listings on dev	Sprint 3 tag	Three services merged	Sprint 4 tag [FILL: date]

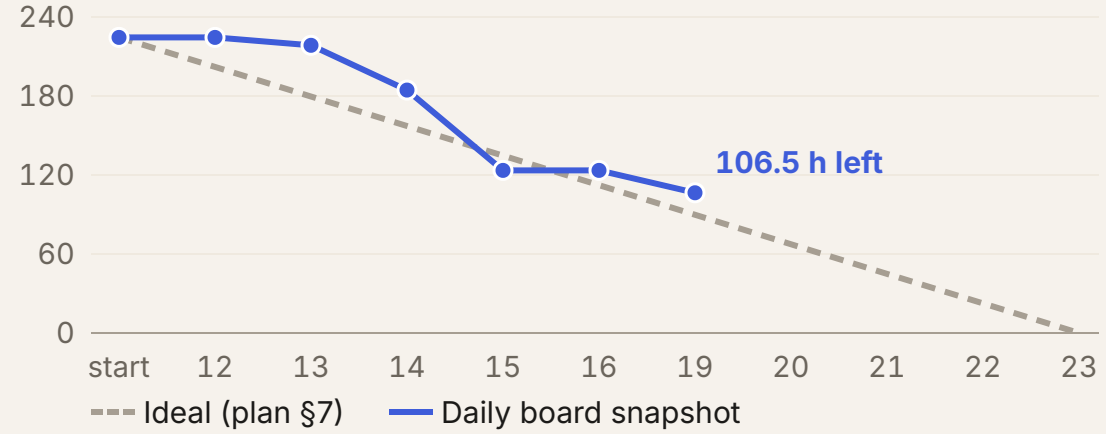
Planned hours per member [VERIFY: S4 split, PR #538]

Member	S2	S3	S4	Board items Done
Liu Jin	30	30	51	39
Yu Qianshuo	30	30	48	8
Wang Ziang	30	30	46	17
Panpakorn	30	30	41.5	29
Lim Jiechao	30	30	38	29

S1 hours were not recorded. Board items are delivery evidence, not effort. Planned S2–S4: 98–111 h each; guideline ≈ 10 man-days each (proposal §7)

[CONFIRM: actual effort per member, agreed by all five]

Sprint 4 burndown to 19 Oct: plan hours not Done



S1		[FILL: S1 hours]
S2		**134** of 150 h
S3		**117** of 150 h
S4		**118** of 224.5 h [FILL: final]

Status, open issues, and what each retro changed

Live on dev · 19 Oct

`[VERIFY: re-check on 25 Oct]`

- API health UP; web app loads
  checked 19 Oct, 14:14 SGT
- 14 listings in the catalog
  startable listings set through the admin API (8 Oct)
- Google sign-in, Secure session cookie
  NFR-01 verified on dev, 8 Oct
- Stripe Checkout opens and returns
  Sandbox, 16 Oct (#517)

Still open

- Webhook never confirms: payments stay “waiting”
  #517 · Yorik
- Scheduling and expiry refunds off on dev
  #223 · Yorik · go/no-go for the live payment demo
- Payment hardening in progress
  #539, #548, #550 · Eric
- Sprint 3 review and retro not yet written
  #451 · Lim

Retro → next sprint

Sprint 1 → 2
No burndown kept → daily board snapshots, hours per story

Sprint 2 → 3
A 10,000-line PR, cliff burndown, owners drifted → PRs under 1,500 lines, mid-sprint check, hours daily

Sprint 3 → 4
`[FILL: retro, #451]` → review: goal not met on dev, so #223 became the critical path

Sprint 4
`[FILL: retro on 23 Oct]`

WZ Eric

Top-down: split by transaction, one codebase, two topologies

[FILL: slide 7 layout]

LJ Liu Jin

UC9 listings: State and Optimistic Offline Lock

[FILL: slide 8 layout]

UC3/UC4 start and join: Factory Method

[FILL: slide 9 layout]

YQ Yorik

UC6 expiry refund: never refund twice

[FILL: slide 10 layout]

Notifications: Observer, Outbox, Idempotent Receiver

[FILL: slide 11 layout]

UC8 My blocs: a frontend design problem

[FILL: slide 12 layout]

LJ Liu Jin

Seven kinds of tests, from domain rules to the deployed site

[FILL: slide 13 layout]

LJ Liu Jin

Gates that block a merge, and what tests caught

[FILL: slide 14 layout]

PS Pan

Source code and artefacts: one trunk, images by digest

[FILL: slide 15 layout]

One pipeline from pull request to dev

[FILL: slide 16 layout]

Security in the pipeline, no stored keys

[FILL: slide 17 layout]

PS Pan

Where it runs: dev on EC2, prod on ECS

[FILL: slide 18 layout]

Technical findings and issues

[FILL: slide 19 layout]

PS Pan

Live: one change from pull request to dev

[FILL: slide 20 layout]

WZ Eric

Live demo: one bloc from invite to refund

[FILL: slide 21 layout]

Thank you

Questions?

github.com/PPsyrius/iss-groupbuy-team8

Backup slides: index

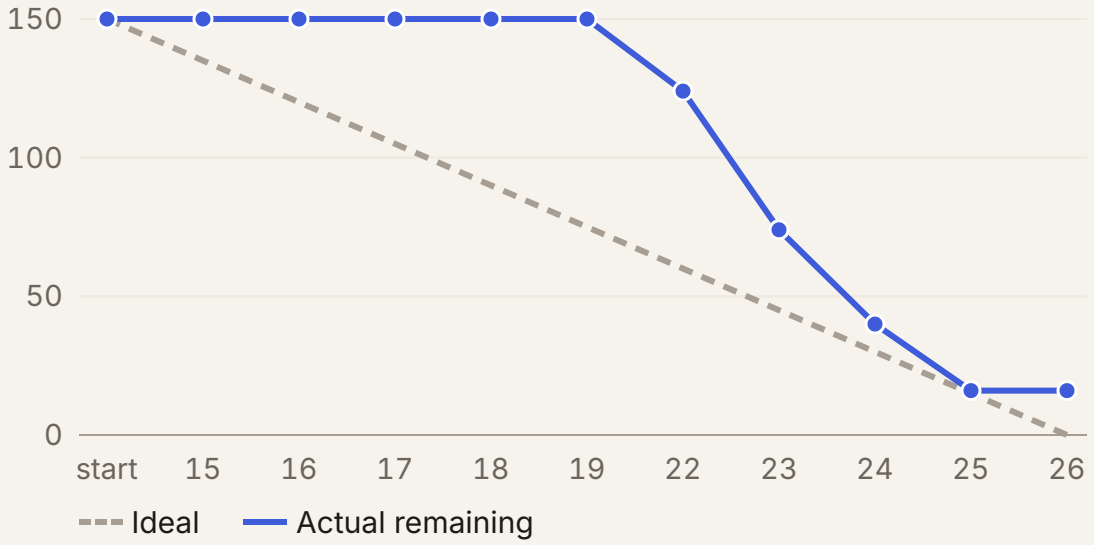
B11 **Sprint 2 and 3 burndowns**
How did earlier sprints burn down?

Click a row (or type its number in the URL as #B6). Slide 22 stays up during Q&A; open a backup only when a question needs it.

Sprint 2 and 3 burndowns

Sprint 2 · remaining hours, 15–26 Sep

150 h planned · 16 h left · reconstructed from merge dates

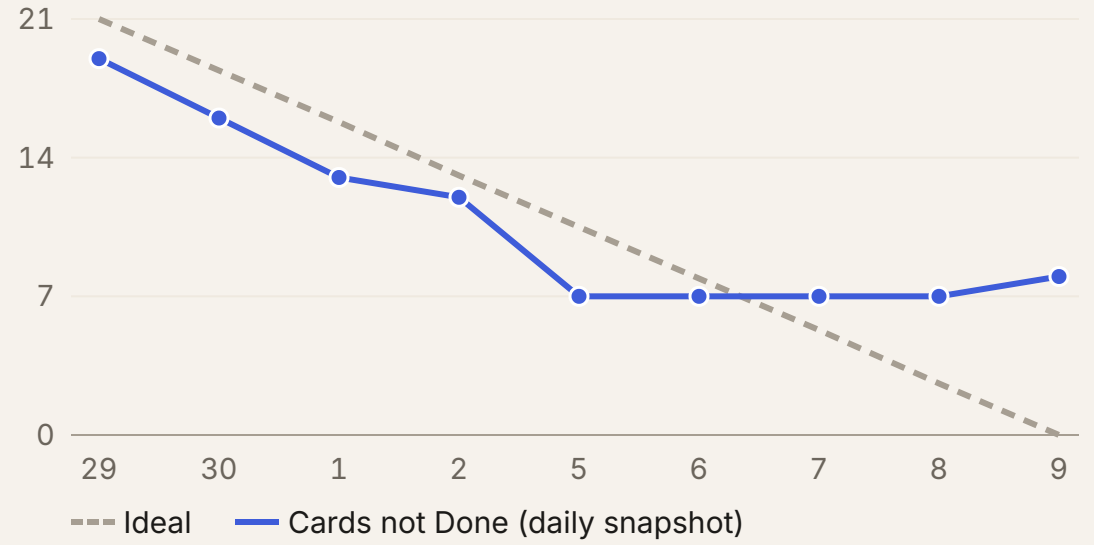


Flat for a week, then a cliff: work merged late in very large pull requests. Retro answer: PRs under 1,500 lines and hours on the board daily.

Source: docs/sprint2-progress-report.typ, "Daily burndown data"

Sprint 3 · open sprint cards, 29 Sep–9 Oct

21 cards · 150 h planned · 33 h left at freeze (review)



Steadier burn, then stuck at 7 from 5 Oct: #223, #224 and #229 waited for Checkout on dev, #226 for the NFR-37 decision. Hours come from the review.

Source: docs/agile/metrics/.csv, sprint-03/review.md*